

Phase 7: Project Documentation

1. System Overview

The Human Development Index (HDI) Prediction System is a Machine Learning-based web application that predicts a country's Human Development Index (HDI) score and classifies it into one of four categories: Very High, High, Medium, or Low Human Development. The application uses socio-economic indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per Capita (PPP). This documentation serves as a guide for both end-users and future developers.

2. End-User Manual

How to Use the Application

Navigate to the deployed web application URL.

On the home page, open the HDI Prediction form.

Enter the required development indicators:

Life Expectancy: Average life expectancy at birth (years).

Mean Years of Schooling: Average years of education completed.

Expected Years of Schooling: Expected years of education for a child entering school.

Gross National Income (GNI) per Capita (PPP): Income per person in PPP (US Dollars).

Click the Predict button.

The system will display:

Predicted HDI Score

HDI Category (Very High, High, Medium, or Low)

A brief explanation of the country's development level

3. Developer Guide

Local Environment Setup

To run this project locally, ensure that Python 3.x is installed.

```
# Install project dependencies
```

```
pip install -r requirements.txt
```

```
# (Optional) Retrain the model if the dataset is updated
```

```
python src/train_model.py
```

Start the Flask application

```
python src/app.py
```

Access the application using:

<http://localhost:5000>

Modifying the Machine Learning Model

To replace or retrain the existing Machine Learning model:

Update the training code in `src/train_model.py`.

Train the new model using the updated dataset.

Save the trained model as `models/hdi_model.pkl`.

Ensure the model loading path in `src/app.py` points to the updated `.pkl` file.

Restart the Flask application and verify that predictions are generated correctly.

Project Structure

HDI_Prediction_System/

|

├── Dataset/

| └── Human Development Index.csv

├── models/

| └── hdi_model.pkl

├── src/

| ├── prepare_dataset.py

| ├── train_model.py

| └── data_analysis.py

| └── app.py

├── templates/

| └── home.html

- | └─ indexnew.html
- └─ static/
- | └─ style.css
- └─ requirements.txt
- └─ README.md

This documentation follows the same format as your sample while being fully customized for your Human Development Index (HDI) Prediction System.